Experiment 1: ER Diagram for Indian E-Commerce Platform

Title:

Design an ER Diagram for an Indian E-Commerce Platform

Aim:

To design an Entity-Relationship (ER) diagram for an Indian e-commerce platform that represents customers, orders, products, sellers, payments, deliveries, addresses, and categories along with their attributes and relationships.

Theory:

An Entity-Relationship (ER) model is a conceptual model used to represent the structure of a database. It shows the important entities of a system, their attributes, and the relationships between them.

In an e-commerce system, entities such as Customer, Order, Product, Seller, Payment, Delivery, Address, and Category are required to store and manage different types of information.

The ER diagram uses different components:

• Entity: Represents a real-world object such as Customer, Product, or Order.

• Attribute: Represents a property of an entity, such as customer name or product price.

• Primary Key (PK): Uniquely identifies each entity record, such as customer_id or product_id.

• Composite Key: A key formed using more than one attribute. In the diagram, ORDER_ITEM uses (order_id, product_id) as a composite key.

• Multi-valued Attribute: An attribute that can contain multiple values, such as phone numbers.

• Composite Attribute: An attribute that can be divided into smaller attributes. For example, Address can contain house number, street, city, state, pincode, and country.

• Relationship: Represents an association between entities, such as Customer places Order.

• Cardinality: Defines the number of entities participating in a relationship, such as 1:N or 1:1.

• Participation Constraint: Defines whether participation of an entity in a relationship is mandatory or optional.

• Specialization (ISA): Divides a general entity into more specific entities. In this diagram, PRODUCT is specialized into PHYSICAL_PRODUCT and DIGITAL_PRODUCT.

DIAGRAM:

